Introduction to Data Analytics and Data Science

Internship Learning Deliverable • Comparison Document & Data Project Lifecycle

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Learning objectives

By the end of this document, the learner should be able to distinguish Data Analytics from Data Science, explain the four common types of analytics, describe how organisations work with data, identify key data roles, and trace how raw data becomes a business decision.

1. Data Analytics vs Data Science

Data Analytics focuses on examining existing data to answer business questions, understand performance, find patterns, and support decisions. It often uses SQL, spreadsheets, dashboards, statistics, and visualisation tools.

Data Science combines statistics, programming, domain knowledge, and machine learning to investigate complex problems, build predictive models, and sometimes create data-driven products. It may use historical data to predict future outcomes.

Aspect	Data Analytics	Data Science
Main purpose	Explain performance and support decisions using data.	Discover patterns and build models or data products.
Typical questions	What happened? Why did sales fall? Which product performs best?	What may happen next? Can we predict demand or detect fraud?
Common methods	Data cleaning, SQL queries, descriptive statistics, dashboards, visualisation.	Statistical modelling, experimentation, machine learning, feature engineering.
Common tools	Excel, SQL, Power BI, Tableau, Python (Pandas).	Python/R, Pandas, scikit-learn, notebooks, ML platforms.
Typical outputs	Reports, KPIs, dashboards, recommendations.	Predictive models, experiments, scoring systems, data products.

Example: A retailer asks why sales decreased last month. A Data Analyst may compare sales by store, product, and date. A Data Scientist may build a model to forecast next month's demand or estimate which customers are likely to stop buying.

2. Four Types of Analytics

Type	Core question	Example
Descriptive	What happened?	A dashboard reports that monthly sales were ■8 lakh.
Diagnostic	Why did it happen?	Analysis finds that sales fell because a popular product was out of stock.
Predictive	What is likely to happen?	A model estimates next month's sales from past sales and seasonal trends.
Prescriptive	What should we do?	The system recommends restocking the product before the next busy period.

Remember the sequence: describe → diagnose → predict → prescribe. Not every project needs all four types.

3. How Organisations Work with Data

1. **Collect:** Gather data from sales systems, websites, mobile apps, surveys, sensors, customer support, and operational databases.
2. **Store:** Keep data in databases, data warehouses, data lakes, or cloud storage with suitable access controls and backups.
3. **Process and clean:** Remove duplicates, handle missing or inconsistent values, standardise formats, and check data quality.
4. **Analyse:** Use queries, statistics, visualisations, and models to find trends, relationships, exceptions, or forecasts.
5. **Communicate:** Present findings through reports, dashboards, charts, and clear explanations for the intended audience.
6. **Decide and monitor:** Choose an action, implement it, measure the result, and repeat the process as new data arrives.

Good practice: protect personal information, restrict access, document assumptions, check data quality, and avoid treating correlation as proof of causation.

4. Key Data Roles

Role	Main responsibilities	Typical outputs
Data Analyst	Clean and query data, calculate metrics, investigate business questions, and explain findings.	Analysis, reports, dashboards, recommendations.
Business Intelligence (BI) Analyst	Define KPIs, build dashboards, model reporting data, and help teams monitor performance.	Interactive dashboards, KPI reports, recurring business insights.
Data Scientist	Use statistics and machine learning to explore complex questions, run experiments, and develop predictive models.	Forecasts, predictive models, experiments, model evaluations.

Data Engineer	Build and maintain reliable data pipelines, integrations, storage, and data infrastructure.	Data pipelines, trusted datasets, data platforms.

These roles overlap in real organisations. The exact responsibilities depend on team size, industry, and the maturity of the data platform.

5. Data Project Lifecycle: From Raw Data to Business Decision

The lifecycle is iterative: results and monitoring often reveal new questions, so teams may return to earlier steps.

1. Define the business problem
Set goal, questions, and success metrics

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2. Collect data
Gather relevant data from approved sources

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3. Store, clean & validate
Fix quality issues and protect sensitive data

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4. Explore & analyse
Use SQL, statistics, charts, and suitable models

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5. Interpret & communicate
Explain findings, limitations, and options

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6. Make a business decision
Choose and implement an evidence-based action

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7. Monitor outcomes & improve
Track KPIs and refine the approach

Worked example — medical store inventory: The owner wants to reduce expired medicines and stock-outs. Collect medicine purchase, sales, expiry, and inventory data; validate product IDs and dates; analyse fast-moving and near-expiry stock; present a dashboard; decide what to reorder or discount according to policy; then monitor expiry losses and availability.

6. Summary

Data Analytics commonly explains business performance and supports decisions. Data Science often applies deeper statistical and machine-learning methods to build predictions or data products. Organisations collect, store, clean, analyse, communicate, and act on data. Analysts, BI Analysts, Data Scientists, and Data Engineers contribute different but connected skills. A useful data project starts with a clear business problem and ends with action and monitoring.